

✓ Homework 4: Supervised Learning

After going through the lecture notebooks, work on the homework below.

Question

Using a dataset from MyAnimeList [1], run a simple linear regression where the target column name is `Y_column` and the explanatory column name is `X_column`, both passed as function arguments, and compute the **coefficient of determination (R-squared)**. Do not transform variables or preprocess data; use the data as-is.

Inputs / Outputs:

```
homework(anime_data_extracted: pd.DataFrame, X_column: str, Y_column: str) -> float
```

Argument	Type	Description
<code>anime_data_extracted</code>	<code>pd.DataFrame</code>	Preprocessed anime dataset (see Section 2)
<code>X_column</code>	<code>str</code>	Explanatory variable column name
<code>Y_column</code>	<code>str</code>	Target variable column name
Return	Type	Description
<code>result</code>	<code>float</code>	Coefficient of determination (R-squared) from simple linear regression

Example:

```
result = homework(anime_data_extracted, X_column='Members', Y_column='Completed')
print(result)
```

```
0.954XXXXXXXXXXXX
```

Submission notes:

- On Omnicampus, submit only the cell marked `# Submit only this cell to Omnicampus`.
- Keep the function name as `homework` and do not change argument names.
- Return the coefficient of determination as a `float`.

Deadline

Wed, May 20 (11:00 UTC)

✓ 1. Importing Libraries

Note: Do not include this section in your `homework()` function.

```
# common
import numpy as np
import pandas as pd
from sklearn import linear_model

# Libraries for retrieving data from the web and handling zip files
import requests, zipfile
from io import StringIO
import io
```

✓ 2. Downloading the Dataset

Download the MyAnimeList dataset from GitHub and clean the data by removing rows where `Score` is `'Unknown'`.

Note: Do not include this section in your `homework()` function.

```
# Specify the url with data
url = 'https://github.com/Hernan4444/MyAnimeList-Database/archive/refs/heads/master.zip'

# Acquire data from the url
r = requests.get(url, stream=True)

# read and extract the zipfile
z = zipfile.ZipFile(io.BytesIO(r.content))
z.extractall()

# Load and clean the data
anime_data = pd.read_csv('MyAnimeList-Database-master/data/anime.csv')
anime_data_extracted = anime_data[anime_data['Score'] != 'Unknown'].copy()
anime_data_extracted['Score'] = pd.to_numeric(anime_data_extracted['Score'])
```

✓ 3. Solution

Implement the function below and submit it to Omnicampus.

```
# Submit only this cell to Omnicampus
def homework(anime_data_extracted, X_column, Y_column):
    #!!WRITE ME!!
    return result
```

✓ 4. Test Your Solution

Run the test cell below to check that your `homework()` function does not contain import statements or data loading code before submitting.

Note: Do not modify the test code.

```
import ast
import inspect

def hw_public_tests():
    print("===HW4 Public Tests ===")

    source = inspect.getsource(homework)
    tree = ast.parse(source)

    # Test 1: Disallow import statements
    print("Test 1: Check for import statements...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, (ast.Import, ast.ImportFrom)):
            print("NG")
            print(" homework() contains import statements. Please remove them.")
            return
    print("OK")

    # Test 2: Disallow download code
    print("Test 2: Check for data downloading code...", end=" ")
    download_keywords = ['requests.get', 'urlopen', 'wget', 'urllib', 'gdown',
                        'source_lower = source.lower()']
    for keyword in download_keywords:
        if keyword in source_lower:
            print("NG")
            print(f" homework() contains data download code: '{keyword}'")
            return
    print("OK")

    # Test 3: Disallow data loading code
    print("Test 3: Check for data loading code...", end=" ")
```

```
loading_keywords = ['read_csv', 'read_excel', 'read_json', 'read_html', 'zi
for keyword in loading_keywords:
    if keyword in source:
        print("NG")
        print(f" homework() contains data loading code: '{keyword}'")
        return
    print("OK")

print("=== All Public Tests Passed ===")

hw_public_tests()
print(homework(anime_data_extracted, X_column='Members', Y_column='Completed'))
```

Once all tests have passed, copy and paste your `homework()` function to Omnicampus and submit your solution. If you see a `1.0`, it means that your answer was correct.

References

[1] Hernan4444. *MyAnimeList Database*. GitHub.

<https://github.com/Hernan4444/MyAnimeList-Database>